

MESSAGE AUTHENTICATION CODES (MACS) FOR AUTHENTICITY/INTEGRITY

Authenticity and Integrity

- ◆ Data privacy is not the only important cryptographic goal
- ◆ It is also important that a receiver is assured that the data it receives has come from the sender and has not been modified on the way (and detect if it is not the case)
- ◆ The goals are data **authenticity** and **integrity**

Encryption Solves Data Privacy, Not Authenticity/Integrity

Recall OneTimePad: $\mathcal{E}(K, M) = K \oplus M$



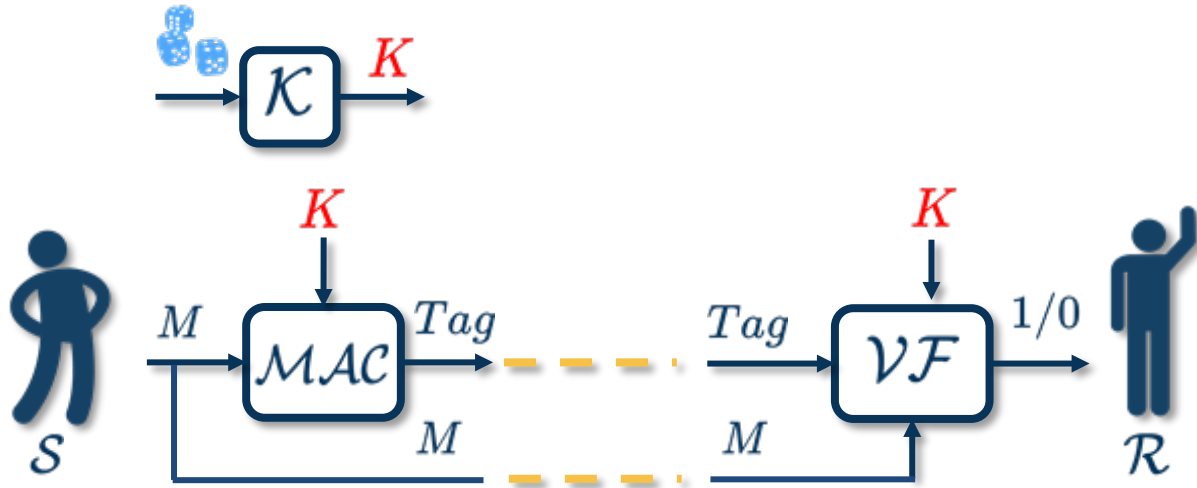
R gets $M \oplus M'$ instead of M

Message Authentication Code (MAC)

- Is the primitive for the goal of data authenticity in the symmetric-key setting
- Is defined by
 - Message space
 - Key generation algorithm (or key space)
 - MAC (tagging, signing) algorithm
 - Verification algorithm

MAC syntax

$\Pi = (\mathcal{K}, \mathcal{MAC}, \mathcal{VF})$ $\mathcal{MsgSp}$ -message space



It is required that for every $M \in \mathcal{MsgSp}$ and every K that can be output by $\mathcal{K}$, $\mathcal{VF}(K, M, \mathcal{MAC}(K, M)) = 1$

Message Authentication Code (MAC)

- ◆ If the key-generation algorithm simply picks a random string from some $KeySp$, then $KeySp$ describes $\mathcal{K}$
- ◆ If the MAC algorithm is deterministic, then the verification algorithm $\mathcal{VF}$ does not have to be defined as it simply re-computes the MAC by invoking the MAC algorithm on the given message M and accepts if the result is equal to its input TAG .

Towards a Security Definition for MACs

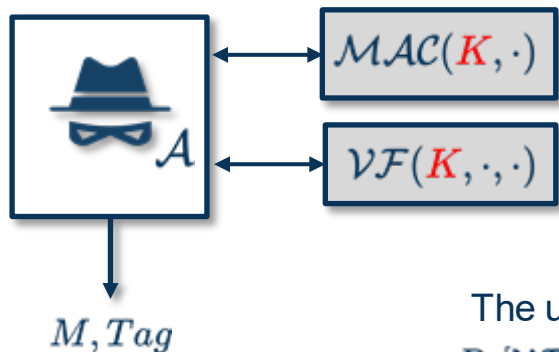
- ◆ We imagine that an adversary can see some number of messages plus tag pairs
- ◆ As usual, it is necessary but not sufficient to require that no adversary can compute the secret key
- ◆ Right now we will not be concerned with replay attacks
- ◆ We don't want an adversary to be able to compute a new message and a tag such that the receiver accepts (outputs 1).

Security Definition for MACs

Fix $\Pi = (\mathcal{K}, \mathcal{MAC}, \mathcal{VF})$

Run $\mathcal{K}$ to get K

For an adversary $\mathcal{A}$ consider an experiment, called uf-cma (unforgeability under chosen message attack)



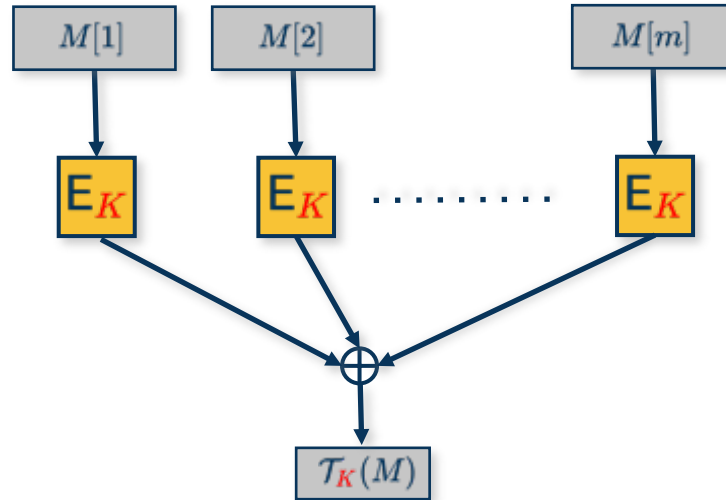
The uf-cma advantage of $\mathcal{A}$, $\text{Adv}^{\text{uf-cma}}(\mathcal{A})$, is defined as $\Pr[\mathcal{VF}(K, M, Tag) = 1 \text{ and } M \text{ was not queried to the } \mathcal{MAC} \text{ oracle}]$

UF-CMA security is defined the usual way.

Example 1

Let $E : \{0, 1\}^k \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a block cipher.

Mac algorithm $\mathcal{T}$

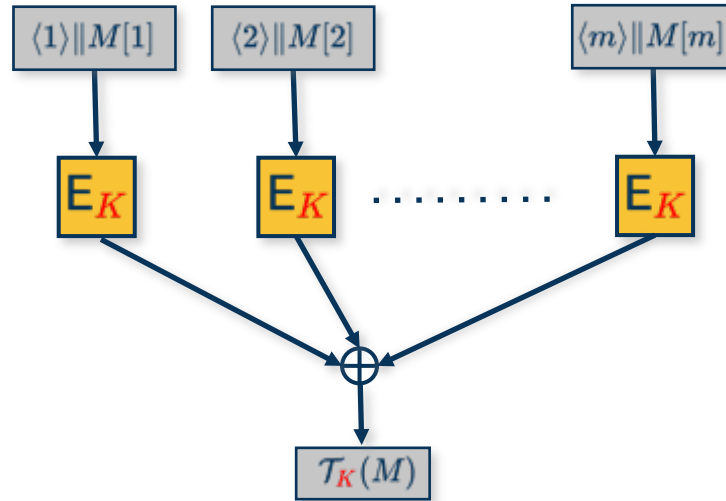


Is this MAC UF-CMA?

Example 2

Let $E : \{0, 1\}^k \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a block cipher.

Mac algorithm $\mathcal{T}$



Notation: $\langle i \rangle$ is a 32-bit representation of the block index i

Notice that the previous attacks fail

Is this MAC UF-CMA?

Weakness

- Weakness: suppose we have

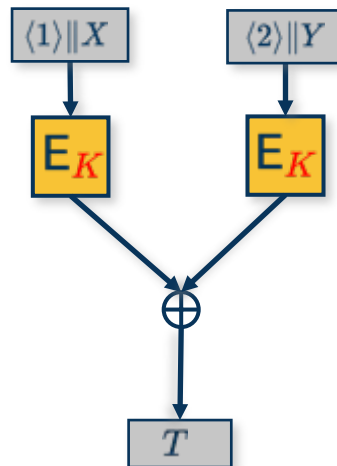
$$T_1 = \mathcal{T}_K(X_1 Y_1) = E_K(\langle 1 \rangle \| X_1) \oplus E_K(\langle 2 \rangle \| Y_1)$$

$$T_2 = \mathcal{T}_K(X_1 Y_2) = E_K(\langle 1 \rangle \| X_1) \oplus E_K(\langle 2 \rangle \| Y_2)$$

$$T_3 = \mathcal{T}_K(X_2 Y_1) = E_K(\langle 1 \rangle \| X_2) \oplus E_K(\langle 2 \rangle \| Y_1)$$

- Add these and we get

$$T_1 \oplus T_2 \oplus T_3 = E_K(\langle 1 \rangle \| X_2) \oplus E_K(\langle 2 \rangle \| Y_2) = \mathcal{T}_K(X_2 Y_2)$$



Note

- ◆ We broke the MAC schemes without breaking the underlying function families (they are secure PRFs)
- ◆ The weaknesses were in the schemes, not the tools

Replay attacks

- ◆ Does the UF-CMA notion capture replay attacks?
 - No. Why?
- ◆ It's better to add replay protection to UF-CMA schemes.
 - How?

PRF as a MAC

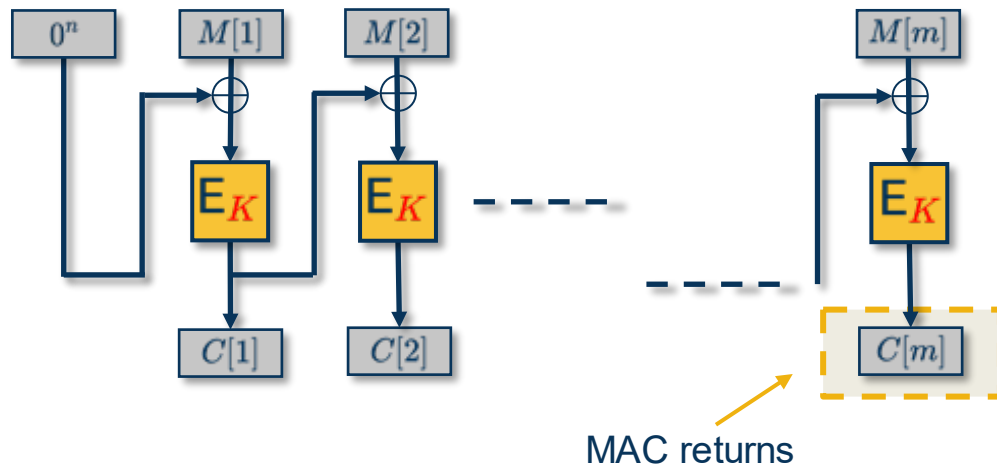
- Any PRF function yields a UF-CMA secure MAC (can be proven)
- But practical PRFs, blockciphers, are only for short inputs
- Can we build blockcipher modes of operation?

CBC-MAC for Fixed-Length Messages

Let $E : \{0, 1\}^k \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a block cipher.

CBC-MAC = $(\{0, 1\}^k, \mathcal{MAC})$:

$\mathcal{MsgSp} = \{0, 1\}^{nm}$ for some $m \geq 1$.



Theorem: For any efficient adversary $\mathcal{A}$ there exists an adversary $\mathcal{B}$ with similar resources such

that $\text{Adv}_{\text{CBC-MAC}}^{\text{uf-cma}}(\mathcal{A}) \leq \text{Adv}_{\text{E}}^{\text{prf}}(\mathcal{B}) + \frac{m^2 q_{\mathcal{A}}^2}{2^{n-1}}$

Other MACs

- CBC-MAC is not UF-CMA for messages or arbitrary (multiple of block length) length messages
- There are several UF-CMA secure blockcipher-based MACs
- Are there MACs based on other primitives, such as hash functions?
 - Yes
 - Let's study hash functions first